Keonwoo Kim

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Research interests

Robotics, Robot Navigation, Foundation Model-based Robotics, Machine learning, Deep Learning, Large Language Model, Vision-Language Action, Policy Learning, RL

Education

2020 – 2027 **Seoul National University** – Seoul, Korea
M.S & Ph.D. Intergrated in Dept. of Electrical and Computer Engineering
Supervised by Prof. Seong-Woo Kim. *GPA: 3.94 / 4.30.*

2015 – 2020 **Hanyang University** – Seoul, Korea
B.S in Dept. of Biomedical Engineering
Supervised by Prof. Anmo Kim. *GPA: 3.89 / 4.50 (Cum Laude)*

Publications

2027 **(TBD) Autonomous Robotic Control System (keywords: VLA, Humanoid)**
Keonwoo Kim, Other ARIL Members, and Seong-Woo Kim.
IEEE International Conference on Robotics and Automation (ICRA), 2027.

2027 **SeDoNa: Self-Determining Robot Navigation in Dynamic Indoor Environments with Spatiotemporal Reasoning**
Keonwoo Kim, Donghwi Jung, Sanghoon Lee, Jiyoon Gong, Seongmin Hong, and Seong-Woo Kim.
IEEE Robotics and Automation Letters (RA-L), 2025.

2025 **LaMOuR: Leveraging Language Models for Out-of-Distribution Recovery in Reinforcement Learning**
Chan Kim, Keonwoo Kim, Mintaek Oh, Seung-Woo Seo,and Seong-Woo Kim.
Submitted to Annual AAAI Conference on Artificial Intelligence (AAAI), 2026.

2025 **E2Map: Experience-and-Emotion Map for Self-Reflective Robot Navigation with Language Models**
(*indicates equal contribution)
Chan Kim*, Keonwoo Kim*, Donghwi Jung, Soojin Woo, Jiyang Lee, Mintaek Oh, Hanbi Baek, Younkyung Woo, John Tucker, Roya Firoozi, Mac Schwager, Seung-Woo Seo, and Seong-Woo Kim
IEEE International Conference on Robotics and Automation (ICRA), 2025.

2025 **GOTPR: General Outdoor Text-based Place Recognition Using Scene Graph Retrieval with OpenStreetMap**
Donghwi Jung, Keonwoo Kim, and Seong-Woo Kim
IEEE Robotics and Automation Letters (RA-L), 2025.

2022 **Let Me Sketch Your Story: An AI Agent That Can Collaborate With Users for Visual Story Generation**
Sihyeon Jo, Keonwoo Kim, Chaeeun Lee, HyeIn Jeon, and Seong-Woo Kim.
Submitted to ACM SIGCHI, 2022.

2022 **Vision-based Collision Avoidance for Mobile Robots through Sim-to-Real Transfer**
Jin-Woo Lee, Keonwoo Kim, Soohyun Shin, and Seong-Woo Kim.
International Conference on Electronics, Information, and Communication (ICEIC), 2022.

2021 **A Robotic Dating Coaching System Leveraging Online Communities Posts**
Sihyeon Jo, Donghwi Jung, Keonwoo Kim, Eun Gyo Joung, Giulia Nespoli, Seungryong Yoo, Minseob So, Seung-Woo Seo, and Seong-Woo Kim.
International Conference on Electronics, Information, and Communication (ICEIC), 2021.

Research experience

January 2022 – November 2023 **Development of Strong Autonomous Intelligence for Unmanned Vehicles in Rough Terrain Missions**
Supervised by Prof. Seong-Woo Kim.
Building autonomous perception module for unmanned vehicles.

March 2022 – December 2022 **Retrieval-Augmented Generation with external domain knowledge for continuous cloud service automation**
Supervised by Prof. Seong-Woo Kim.
Building an open-domain question-answering chatbot system with real-field text corpus from LG Electronics

July 2020 – December 2021

R&D Project for AI for Humanist

Supervised by Prof. Seong-Woo Kim.

Construct text-to-image retrieval system using Vision-and-Language multimodal neural networks.

Teaching experience

Spring 2022 **Global teaching assistant, MMU 07202: Sensors and Actuators (Arusha Technical College, Tanzania)**
Teaching assistant, arranged special lab session (Topic: Audio and Speech Sensing).

Winter 2021 **Engineering mentor, Machine Learning Engineer Procedure (SK Hynix)**
Engineering mentor, advising personal machine learning projects of SK Hynix engineers.

Fall 2021 **Teaching assistant, M2866.003700 001: Autonomous Robot Intelligence (Seoul National University)**
Head teaching assistant, arranged lab session for Natural Language Processing (Topic: Neural Machine Translation).

Spring 2021 **Teaching assistant, M2866.004300 001: Artificial Intelligence for Manufacturing (Seoul National University)**
Head teaching assistant, arranged lab session for Generative Model (Topic: Image-to-Image Translation).

Patents

December 2024 — Keonwoo Kim, Sihyeon Jo, and Seong-Woo Kim, "Emotion Reflective Cost Map based Autonomous Mobile Navigation System," applied for a patent (10-2024-0176918)

October 2021 — Keonwoo Kim, Sihyeon Jo, and Seong-Woo Kim, "Sentence-Based Sketch Recommendation System And Method," applied for a patent (10-2021-0147002)

October 2021 — Sihyeon Jo, Keonwoo Kim, and Seong-Woo Kim, "Sequential Image Retrieval System Utilizing Graph Similarities," applied for a patent (10-2021-0147034)

Technical skills

Programming skills
Python (PyTorch, TensorFlow 2.0, LangChain, etc), C/C++, MATLAB, CSS

Robotics
ROS1, ROS2, CARLA, Isaac Sim, Gazebo

Miscellaneous
LaTeX, Git